2010 Team Problems

T-1. Compute all ordered pairs of real numbers (x, y) that satisfy both of the equations:

$$x^2 + y^2 = 6y - 4x + 12 \qquad \text{and} \qquad 4y = x^2 + 4x + 12.$$

T-2. Define $\log^*(n)$ to be the smallest number of times the log function must be iteratively applied to n to get a result less than or equal to 1. For example, $\log^*(1000) = 2$ since $\log 1000 = 3$ and $\log(\log 1000) = \log 3 = 0.477\ldots \leq 1$. Let a be the smallest integer such that $\log^*(a) = 3$. Compute the number of zeros in the base 10 representation of a.

T-3. An integer N is worth 1 point for each pair of digits it contains that forms a prime in its original order. For example, 6733 is worth 3 points (for 67, 73, and 73 again), and 20304 is worth 2 points (for 23 and 03). Compute the smallest positive integer that is worth exactly 11 points. [Note: Leading zeros are <u>not</u> allowed in the original integer.]

T-4. The six sides of convex hexagon $A_1A_2A_3A_4A_5A_6$ are colored red. Each of the diagonals of the hexagon is colored either red or blue. Compute the number of colorings such that every triangle $A_iA_jA_k$ has at least one red side.

T-5. Compute the smallest positive integer n such that n^n has at least 1,000,000 positive divisors.

T-6. Given an arbitrary finite sequence of letters (represented as a word), a *subsequence* is a sequence of one or more letters that appear in the same order as in the original sequence. For example, N, CT, OTT, and $CONTEST$ are subsequences of the word $CONTEST$, but NOT, $ONSET$, and $TESS$ are not. Assuming the standard English alphabet $\{A, B, \ldots, Z\}$, compute the number of distinct four-letter "words" for which EE is a subsequence.

T-7. Six solid regular tetrahedra are placed on a flat surface so that their bases form a regular hexagon $\mathcal{H}$ with side length 1, and so that the vertices not lying in the plane of $\mathcal{H}$ (the "top" vertices) are themselves coplanar. A spherical ball of radius r is placed so that its center is directly above the center of the hexagon. The sphere rests on the tetrahedra so that it is tangent to one edge from each tetrahedron. If the ball's center is coplanar with the top vertices of the tetrahedra, compute r.

T-8. Derek starts at the point $(0, 0)$, facing the point $(0, 1)$, and he wants to get to the point $(1, 1)$. He takes unit steps parallel to the coordinate axes. A move consists of either a step forward, or a 90° right (clockwise) turn followed by a step forward, so that his path does not contain any left turns. His path is restricted to the square region defined by $0 \leq x \leq 17$ and $0 \leq y \leq 17$. Compute the number of ways he can get to $(1, 1)$ without returning to any previously visited point.

T-9. The equations $x^3 + Ax + 10 = 0$ and $x^3 + Bx^2 + 50 = 0$ have two roots in common. Compute the product of these common roots.

T-10. Points A and L lie outside circle ω, whose center is O, and $\overline{AL}$ contains diameter $\overline{RM}$, as shown below. Circle ω is tangent to $\overline{LK}$ at K. Also, $\overline{AK}$ intersects ω at Y, which is between A and K. If $KL = 3$, $ML = 2$, and $\text{m}\angle AKL - \text{m}\angle YMK = 90^\circ$, compute $[AKM]$ (i.e., the area of $\triangle AKM$).

